

STUDENT PERFORMANCE TRACKER MODEL

1. Introduction

The Student Performance Tracker is a Machine Learning project developed to predict the final academic grade of students using their academic and personal information. It analyzes historical student data to estimate final performance.

2. Objective

- To analyze student academic performance using Machine Learning.
- To preprocess categorical and numerical student data.
- To predict the final grade of a student.
- To compare actual student scores with predicted scores.
- To evaluate the performance of the trained model.
- To provide a simple approach for tracking student academic performance.

3. Dataset & Preprocessing

The Student Performance dataset is obtained from the UCI Machine Learning Repository. The dataset contains **649 student records and 30 input features**, including age, gender, study time, failures, family support, internet availability, absences, and other academic and personal information.

The final grade **G3 (0-20)** is used as the target variable. Categorical features are converted into numerical values using Label Encoding. The dataset is divided into **80% training data and 20% testing data**.

4. Code Implementation

```
import pandas as pd
from ucimlrepo import fetch_ucirepo
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_absolute_error, r2_score

student = fetch_ucirepo(id=320)

X = student.data.features.copy()
y = student.data.targets["G3"]

print(X.head())
print("Dataset Shape:", X.shape)

for col in X.select_dtypes(include="object").columns:
    le = LabelEncoder()
    X[col] = le.fit_transform(X[col])
```

```

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

model = RandomForestRegressor(n_estimators=100, random_state=42)

model.fit(X_train, y_train)

pred = model.predict(X_test)

print("MAE:", mean_absolute_error(y_test, pred))
print("R2 Score:", r2_score(y_test, pred))

result = pd.DataFrame({
    "Actual Score": y_test.values,
    "Predicted Score": pred.round(2)
})

print(result.head(10))

```

Output

Dataset Shape: 649 rows x 30 features

Mean Absolute Error (MAE): 2.0828

R² Score: 0.1591

Student	Actual Score	Predicted Score
1	19	12.24
2	12	12.27
3	18	12.99
4	11	12.08
5	11	12.01
6	17	12.73
7	18	14.57
8	8	8.98
9	10	12.06
10	11	11.98

5. Machine Learning Model

The **Random Forest Regressor** is used to predict the final student grade. It combines multiple Decision Trees to generate the final prediction and reduce overfitting.

The model uses **100 decision trees** and learns patterns between the 30 student features and the final

grade G3.

6. Model Evaluation

The model achieved a **Mean Absolute Error (MAE) of 2.08**, which means its predictions differ from the actual grades by approximately 2.08 marks on average.

The **R² Score is 0.159**, indicating that the model explains about **15.9% of the variation** in final student grades. This shows that the available features provide limited predictive power for accurately estimating the final grade.

7. Limitations

- The model achieved a relatively low R² Score, indicating limited prediction capability.
- Student performance may depend on important academic factors not included in the available input features.
- The dataset represents a limited student population and may not generalize to all educational institutions.

8. Future Scope

- More academic features such as previous exam scores, assignments, and attendance percentage can be included.
- Different Machine Learning algorithms can be tested and compared for better prediction accuracy.
- A web-based dashboard can be developed for student performance tracking and analysis.

9. Conclusion

The Student Performance Tracker uses a **Random Forest Regressor** to predict final student grades using 649 student records and 30 input features.

The model obtained an **MAE of 2.08 and an R² Score of 0.159**. The project successfully demonstrates dataset preprocessing, model training, prediction, and evaluation for student performance analysis.